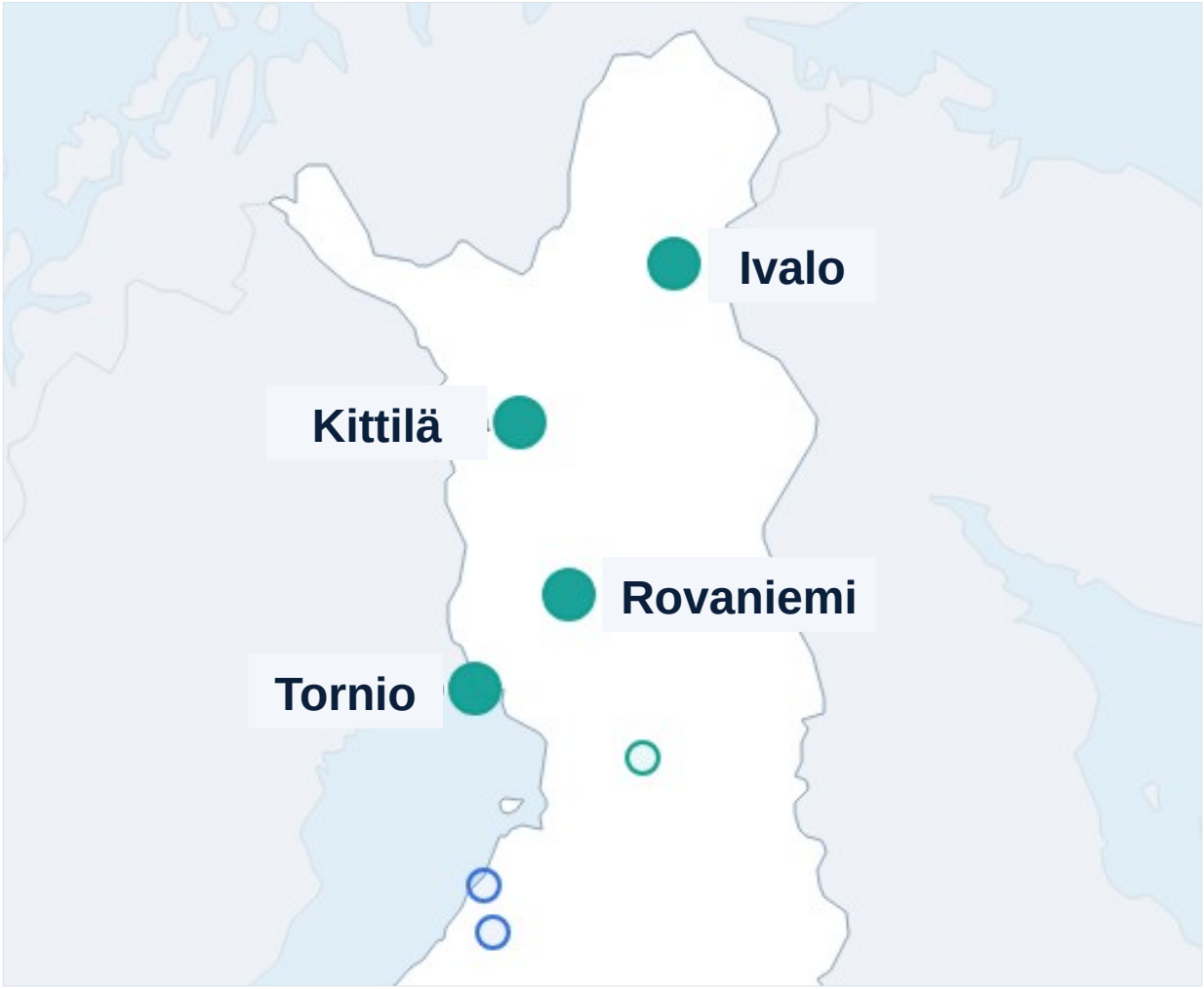
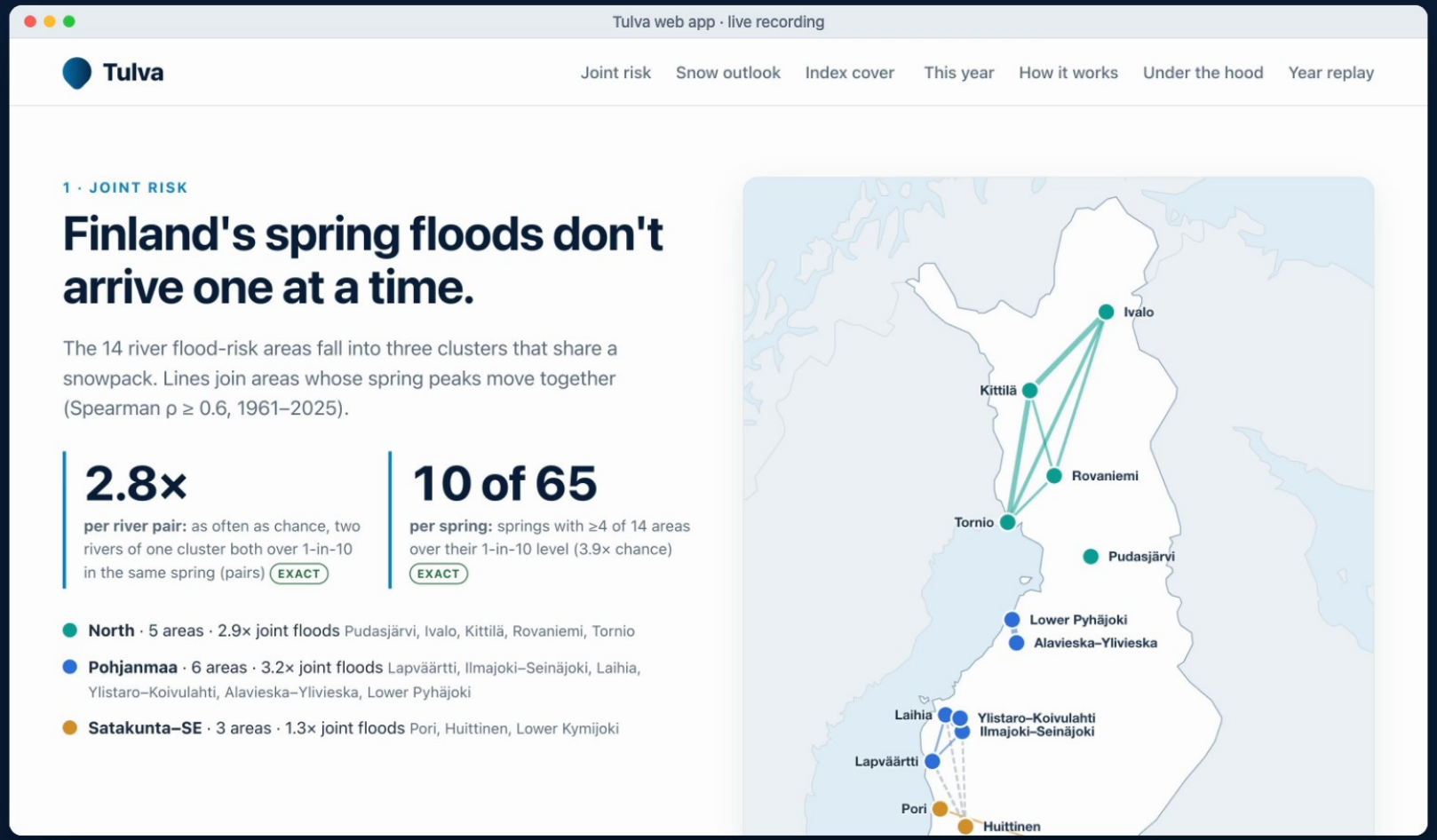
Tulva

May 2005: four Lapland rivers flooded together.

The March snow warned us.

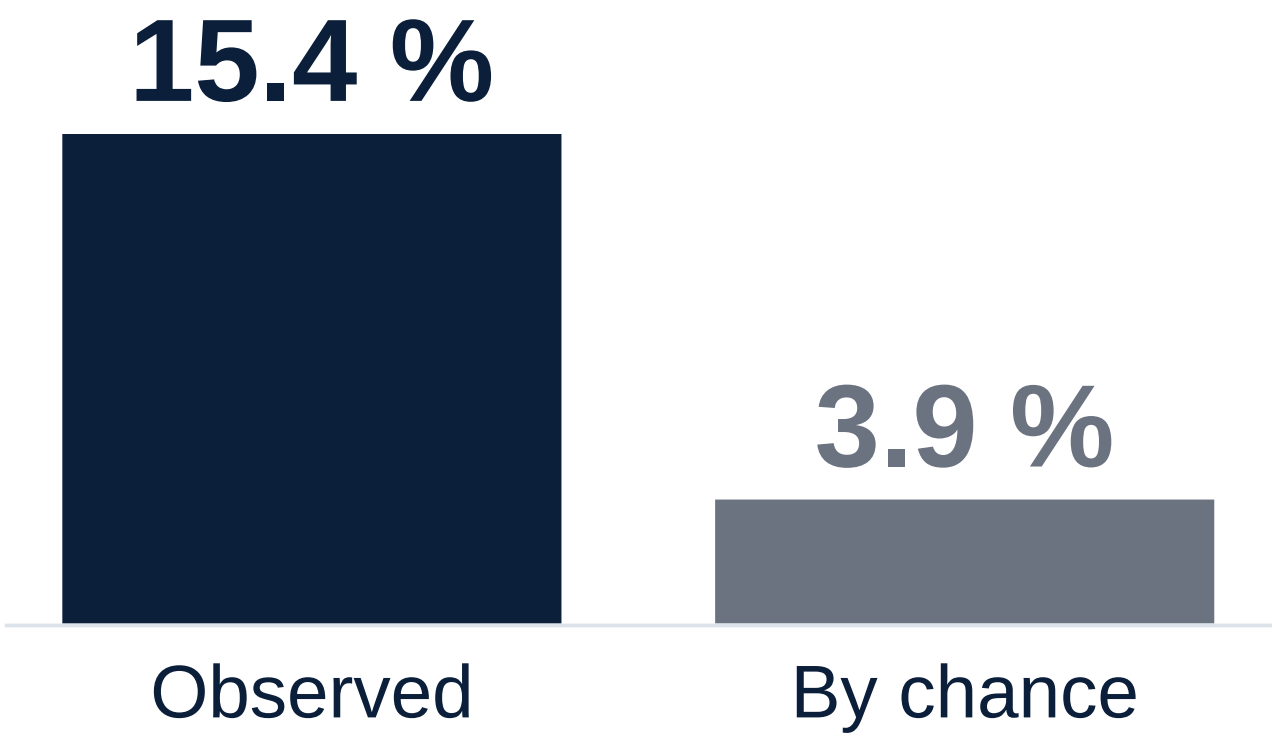




Our app turns that snow into a price.

Rivers flood together, not one at a time

Springs when ≥4 of 14 rivers flood

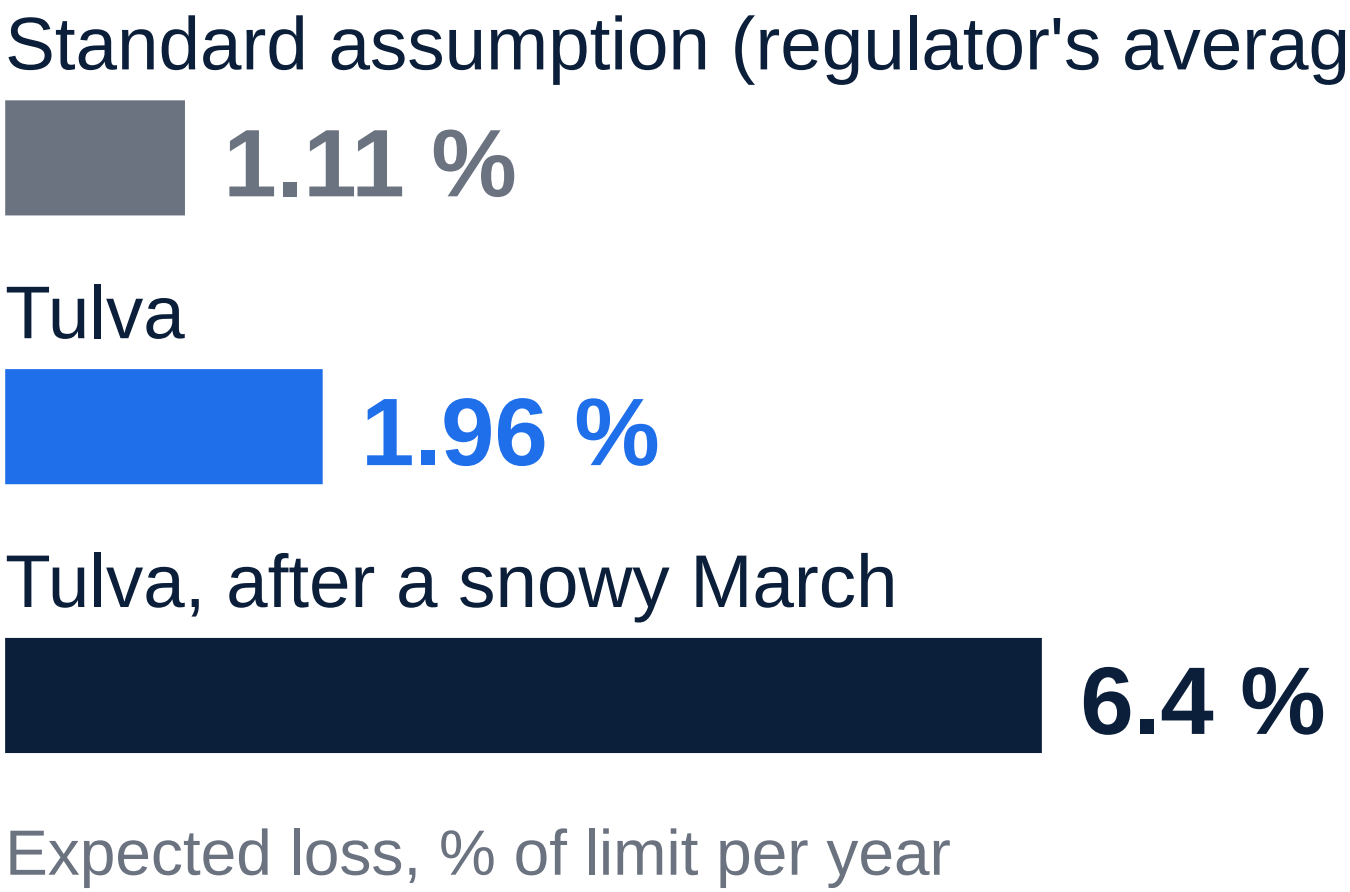


≈4×

as often as chance

Cover is bought in autumn, before the snow.

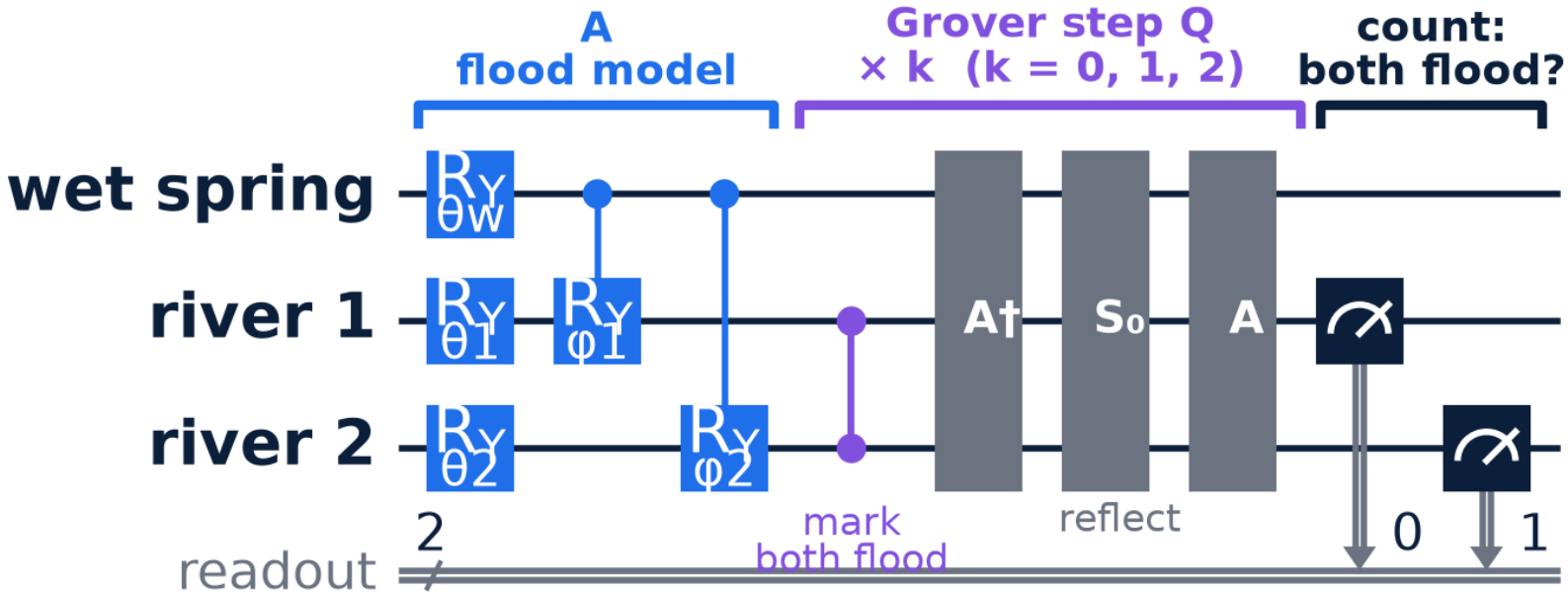
Tulva Index: pays if ≥2 northern rivers flood



×3

chance of a big joint flood after a snowy March

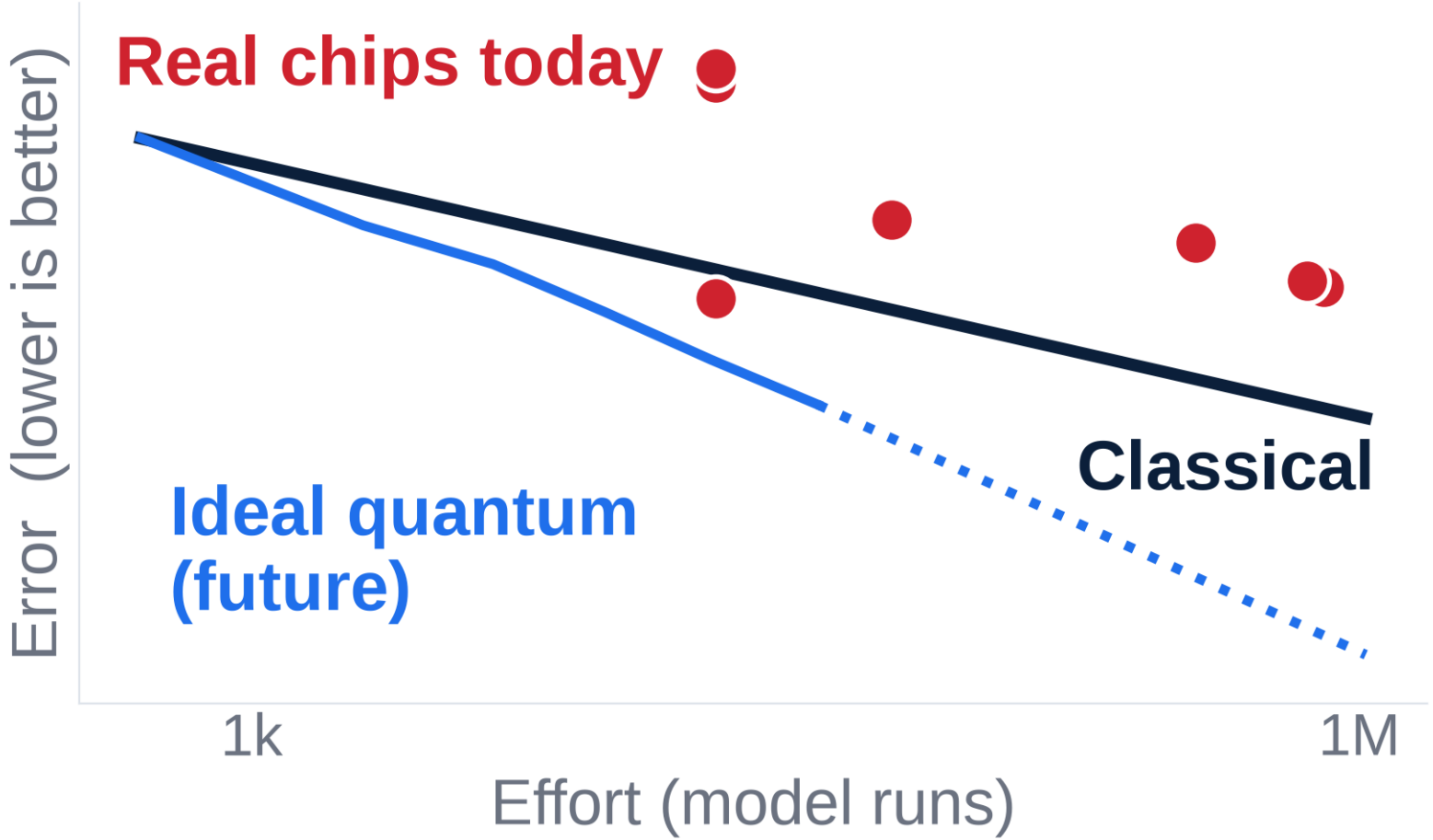
Our circuit: quantum amplitude estimation



Qubits = wet spring + rivers. Gates = our flood model.

Why quantum? 2× accuracy: sampling needs 4× the runs, QAE 2×.

Classical still wins today



Rare 1-in-50 floods drown in noise.

Prices are computed classically.

Who pays



€25k pilot, then €75k / yr

Year 3: €0.2–0.45M / yr

3–6 Finnish insurers, then the Nordics

Ask: 8 weeks of one insurer's claims data, to test on real losses.

Built during this hackathon

- SYKE data pipeline
- Joint flood model
- Index pricing
- Live app
- Runs on 4 quantum chips

Next March, the snow will warn us again.

This time, it will have a price.

Toivo Kallio · **Vaishnav Rane**

Team 21 STUHI · Quantum x Finance 2026